

NEOBIC-VR

Self-Observation Virtual Reality Framework for Physical Exercise in Oncological Care

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Motivation

1. Cancer treatments often lead to extreme fatigue, muscle loss, and reduced quality of life.

2. Patients drop-off → Exercise can improve strength, endurance, mood, and anxiety, **but**:

- Low self-efficacy: patients doubt their capacity to perform and progress
- Monotony of traditional exercise reduces engagement

If I See *Me*, I Believe in *Me*.

Architecture

SELF-AVATAR SELECTION

EMBODIMENT

EXERCISE EXECUTION

INTENSITY CHOICE

SELF-OBSERVATION

Every participant has their **own self-avatar**. Choosing an avatar that matches your appearance leverages self-similarity to deepen the **sense of ownership** and presence during VR sessions.

The participant executes some **embodiment exercises** before entering the exercise session. Seeing your virtual body move in real time via a mirror boosts the sense that “this body is mine”.

The participant performs the exercise three times, with a resting time between each round.

Seeing “you” succeed first makes you more confident and likely to replicate the movement correctly and with higher intensity.

Before performing the exercises, a **twin-avatar** of the participant demonstrates how to perform the exercise at different intensity levels.

Methodology

PARTICIPANTS

N=300

1. VR Exercise / Self-Observation

2. VR Exercise

3. Non-VR Exercise

4. No Exercise

BASELINE CHARACTERIZATION

- Collect a comprehensive oncological profile via the ECOG instrument.
- Assess physical activity levels with the IPAQ and baseline fatigue (e.g., FACIT-Fatigue).
- Measure well-being using validated quality-of-life scales (e.g., EORTC QLQ-C30).
- Self-efficacy (e.g., Marcus et al, 1992)

INTERVENTION PROTOCOL

- Groups 1, 2, and 3 complete the same four-exercise VR regimen over 4 weeks.
- Group 1 watches their personalized avatar demonstrate each exercise sequence before execution.
- Group 3 gets the same exercise routine recommendation but without VR.

Expected Results

	Exercise Adherence	Fatigue Reduction	Self-Efficacy	QoL
Group 1	≥ 75 % of prescribed sessions completed	Mean decrease ≥ 6 points	+ 1.2-point gain	Mean increase 12-15 points
Group 2	~ 60 % completion	Mean decrease ~ 4 points	0.8-point gain	Mean increase 8-10 points
Group 3	≤ 50 % completion	Mean decrease ≤ 2 points	0.3-point gain	Mean increase 8-10 points

References & Contact

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